

Lecture11: Classification3

Naïve Bayes

STA 4724

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Probability Prediction

- In the previous section, we used logistic regression to estimate the probability that a data instance belongs to a particular class and decide the label it should be assigned to
- We were predicting the probability for event A to occur
 - Event A = the data instance belongs to class/label A
 - Denote $\mathbb{P}(A)$

Frequentist and Bayesian

There are two probability schools of thought: frequentist and Bayesian

- From a traditional standpoint, probability is presented in terms of a **frequentist** view where data instances are drawn from a repeatable random sample with parameters that remain constant during the repeatable process.
 - Probability determines the frequency an event occurs
- The Bayesian view takes the approach that data instances are observed from a realized sample, and that the parameters are unknown. Since repeatability of the data sample does not hold under this view, the Bayesian probability is not given in terms of frequencies, but instead it represents a state of knowledge or a state of “belief”.

Bayes' Theorem

- Bayes' theorem states that the probability of a particular hypothesis is given by both current information (data) and prior knowledge.
 - The prior information may be the outcome of earlier experiments or trials, or even educated guesses drawn from experience.
- Let us consider the set of all possible events Ω , which is our sample space. Event A is a member of the sample space
- The probability of the sample space is $\mathbb{P}(\Omega) = 1$. Thus, using $| \cdot |$ denotes the cardinality

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}$$

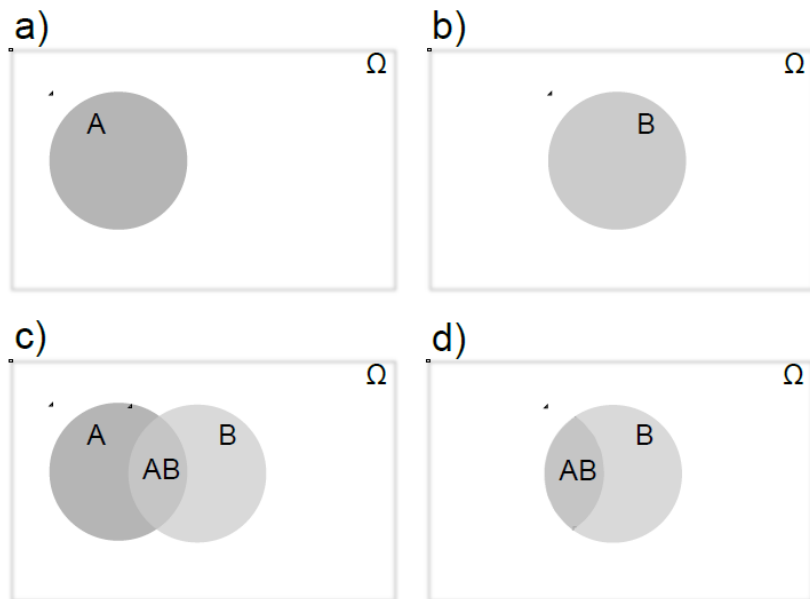
Bayes' Theorem

- Similarly,

$$\mathbb{P}(B) = \frac{|B|}{|\Omega|}, \mathbb{P}(A \cap B) = \frac{|A \cap B|}{|\Omega|}$$

- Thus,

$$\mathbb{P}(A|B) = \frac{|A \cap B|}{|B|} = \frac{\frac{|A \cap B|}{|\Omega|}}{\frac{|B|}{|\Omega|}} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$



Bayes' Theorem

- Same calculation applies to $\mathbb{P}(B|A)$

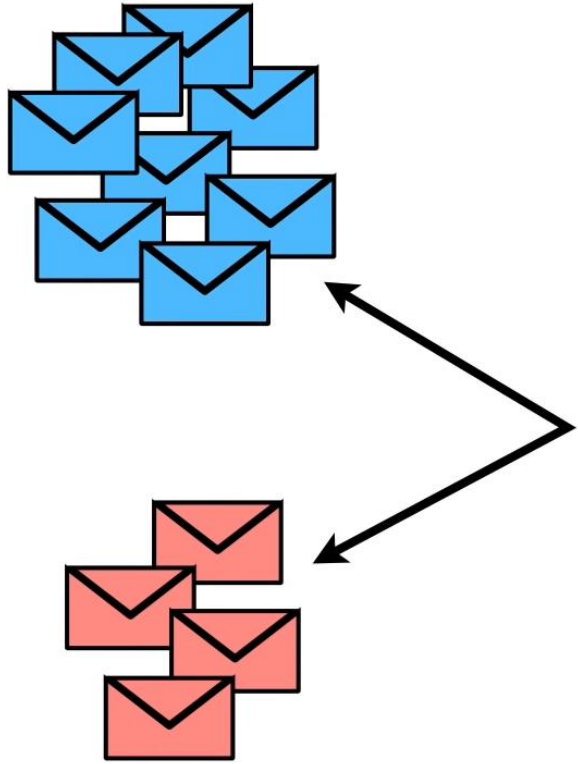
$$\mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(A \cap B) = \mathbb{P}(B|A)\mathbb{P}(A)$$

- Finally, the famous Bayes' Theorem:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}$$

- Naïve Bayes is a classifier based on the idea of Bayes' theorem.

Toy Example: Naïve Bayes



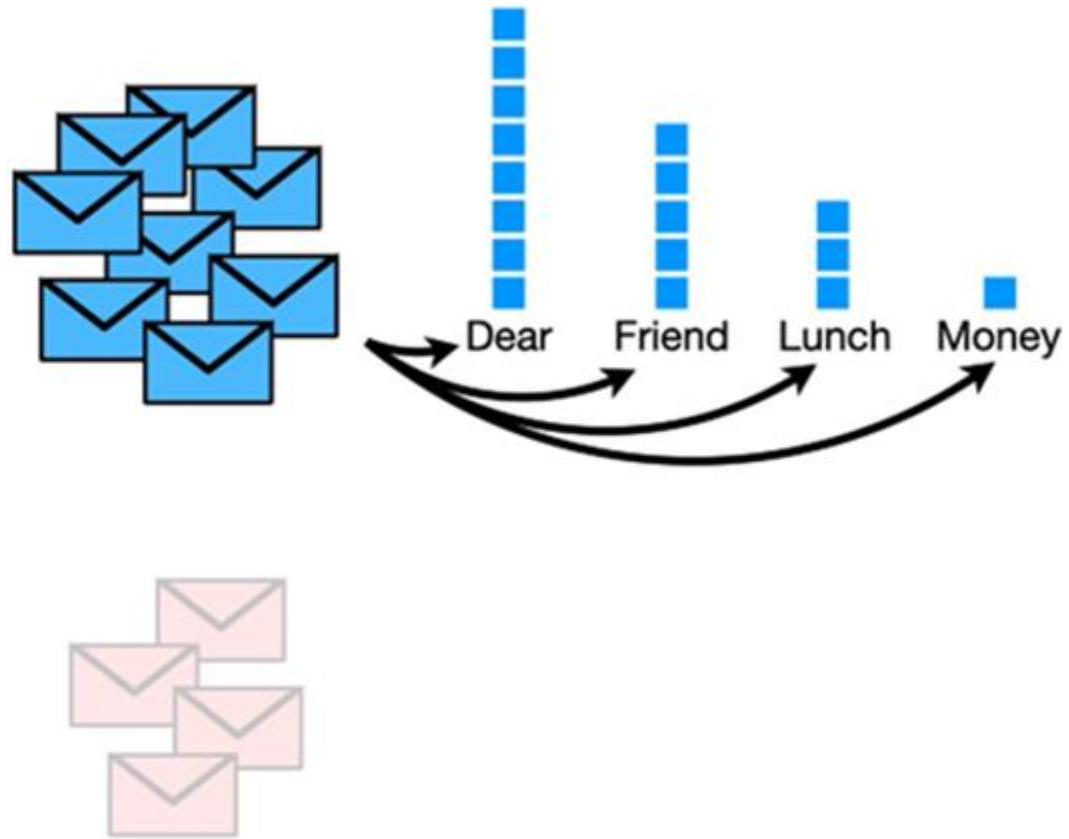
- A well-known application for naïve bayes is spam message filter
- Let $\{x_i\}$ be the content of the message

$$\mathbb{P}(Normal|\{x_i\}) = \frac{\mathbb{P}(\{x_i\}|Normal)\mathbb{P}\{Normal\}}{\mathbb{P}(\{x_i\})}$$

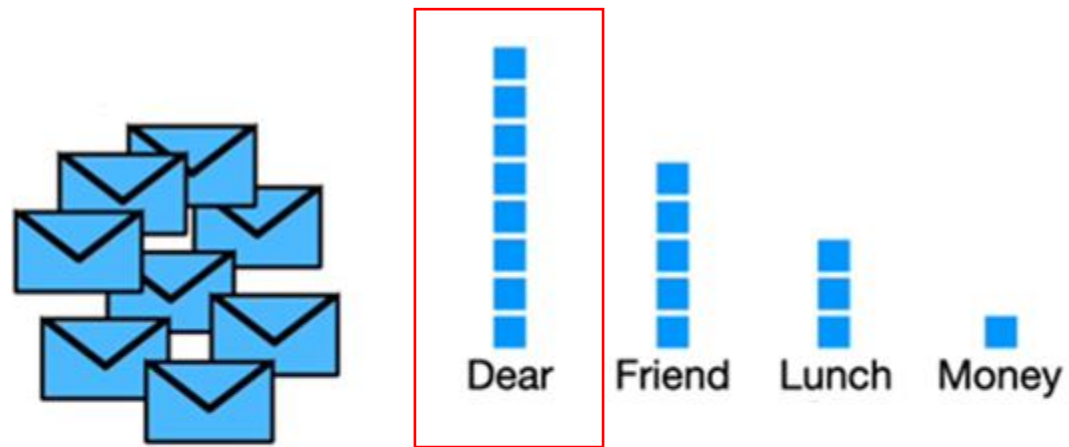
$$\mathbb{P}(Spam|\{x_i\}) = \frac{\mathbb{P}(\{x_i\}|Spam)\mathbb{P}\{Spam\}}{\mathbb{P}(\{x_i\})}$$

- Note that $\mathbb{P}(\{x_i\})$ is a constant that does not depend on the classes *Normal* and *Spam*. It can be incorporated at the end of all our computations.

Toy Example: Naïve Bayes



Toy Example: Naïve Bayes

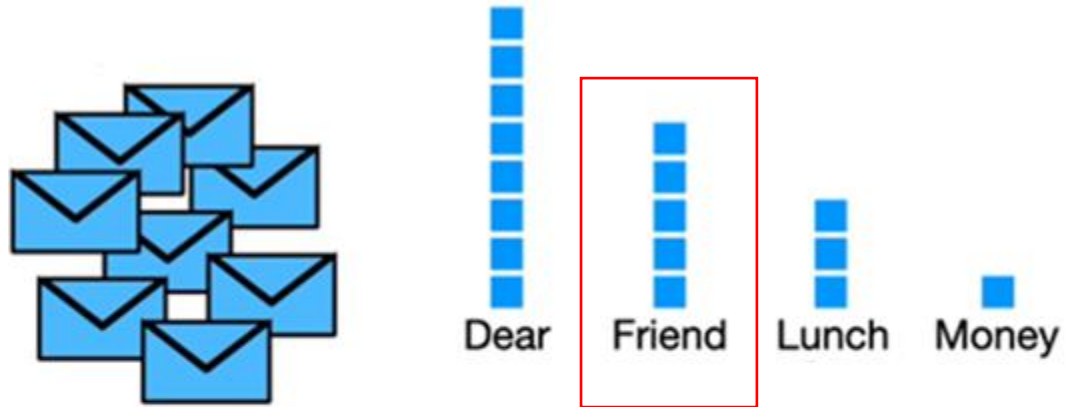


$$\mathbb{P}(\text{Dear}|\text{Normal}) = 8/17$$



Toy Example: Naïve Bayes

$$\mathbb{P}(\text{Dear}|\text{Normal}) = 0.47$$



$$\mathbb{P}(\text{Friend}|\text{Normal}) = 5/17$$

Toy Example: Naïve Bayes

$$\mathbb{P}(\textit{Dear}|\textit{Normal}) = 0.47$$



Dear

$$\mathbb{P}(\textit{Friend}|\textit{Normal}) = 0.29$$



Friend

$$\mathbb{P}(\textit{Lunch}|\textit{Normal}) = 0.18$$



Lunch

$$\mathbb{P}(\textit{Money}|\textit{Normal}) = 0.06$$

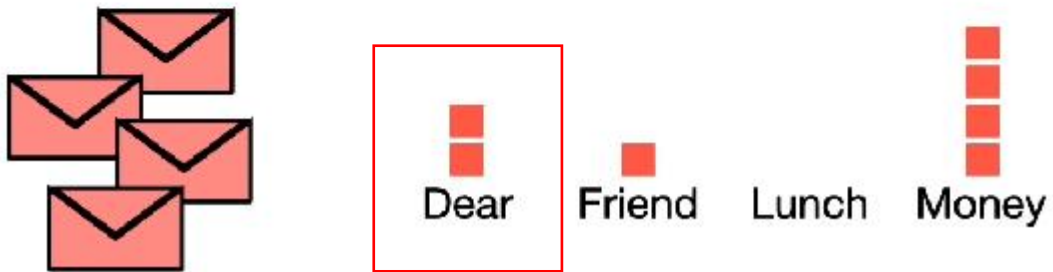
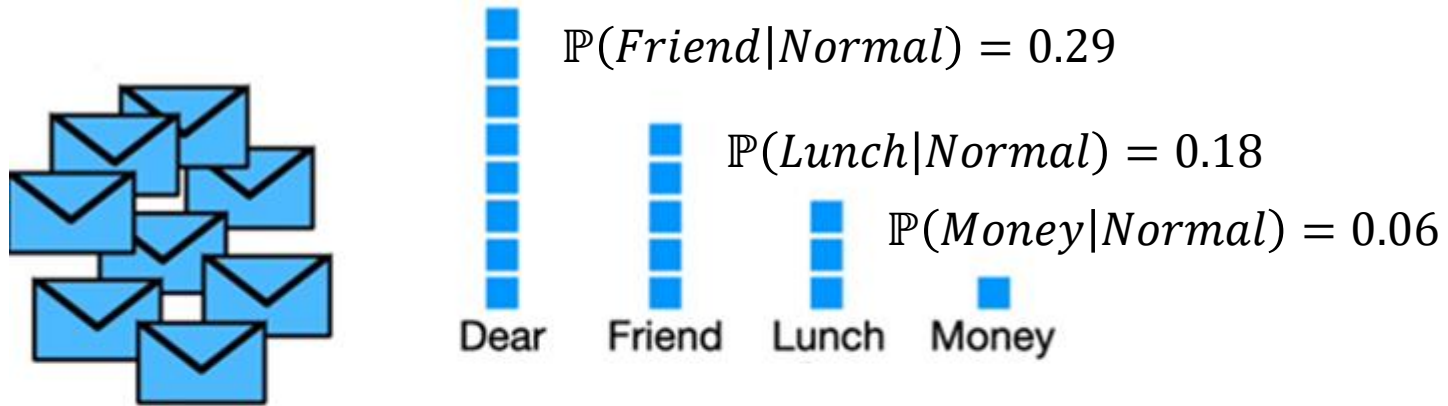


Money



Toy Example: Naïve Bayes

$$\mathbb{P}(\text{Dear}|\text{Normal}) = 0.47$$



$$\mathbb{P}(\text{Dear}|\text{Spam}) = 2/7$$

Toy Example: Naïve Bayes



$$\mathbb{P}(\textit{Dear}|\textit{Normal}) = 0.47$$

$$\mathbb{P}(\textit{Friend}|\textit{Normal}) = 0.29$$

$$\mathbb{P}(\textit{Lunch}|\textit{Normal}) = 0.18$$

$$\mathbb{P}(\textit{Money}|\textit{Normal}) = 0.06$$



$$\mathbb{P}(\textit{Dear}|\textit{Spam}) = 0.29$$

$$\mathbb{P}(\textit{Friend}|\textit{Spam}) = 0.14$$

$$\mathbb{P}(\textit{Lunch}|\textit{Spam}) = 0.00$$

$$\mathbb{P}(\textit{Money}|\textit{Spam}) = 0.57$$

Toy Example: Naïve Bayes



$$\mathbb{P}(\text{Dear}|\text{Normal}) = 0.47$$

$$\mathbb{P}(\text{Friend}|\text{Normal}) = 0.29$$

$$\mathbb{P}(\text{Lunch}|\text{Normal}) = 0.18$$

$$\mathbb{P}(\text{Money}|\text{Normal}) = 0.06$$



$$\mathbb{P}(\text{Dear}|\text{Spam}) = 0.29$$

$$\mathbb{P}(\text{Friend}|\text{Spam}) = 0.14$$

$$\mathbb{P}(\text{Lunch}|\text{Spam}) = 0.00$$

$$\mathbb{P}(\text{Money}|\text{Spam}) = 0.57$$

- Now, a new letter say “Dear Friend” comes in. We want to figure out if it is a normal message or a spam.

- $\mathbb{P}(\text{Normal}) = \frac{8}{8+4} = 0.67$ Because 8 normal messages and 4 spams in the training set

- $\mathbb{P}(\text{Normal}) * \mathbb{P}(\text{Dear}|\text{Normal}) * \mathbb{P}(\text{Friend}|\text{Normal}) = 0.09$
 - This is the score for “dear Friend” being a normal message

- $\mathbb{P}(\text{Spam}) * \mathbb{P}(\text{Dear}|\text{Spam}) * \mathbb{P}(\text{Friend}|\text{Spam}) = 0.01$

- $0.09 > 0.01$, so this is likely a normal message

Toy Example: Naïve Bayes



$$\mathbb{P}(\text{Dear}|\text{Normal}) = 0.47$$

$$\mathbb{P}(\text{Friend}|\text{Normal}) = 0.29$$

$$\mathbb{P}(\text{Lunch}|\text{Normal}) = 0.18$$

$$\mathbb{P}(\text{Money}|\text{Normal}) = 0.06$$

- $\mathbb{P}(\text{Normal}) * \mathbb{P}(\text{Dear}|\text{Normal}) * \mathbb{P}(\text{Friend}|\text{Normal})$ is proportional to $\mathbb{P}(\text{Normal}|\text{Dear Friend})$

- $\mathbb{P}(\text{Spam}) * \mathbb{P}(\text{Dear}|\text{Spam}) * \mathbb{P}(\text{Friend}|\text{Spam})$ is proportional to $\mathbb{P}(\text{Spam}|\text{Dear Friend})$



$$\mathbb{P}(\text{Dear}|\text{Spam}) = 0.29$$

$$\mathbb{P}(\text{Friend}|\text{Spam}) = 0.14$$

$$\mathbb{P}(\text{Lunch}|\text{Spam}) = 0.00$$

$$\mathbb{P}(\text{Money}|\text{Spam}) = 0.57$$

Toy Example: Naïve Bayes



$$\mathbb{P}(\textit{Dear}|\textit{Normal}) = 0.47$$

$$\mathbb{P}(\textit{Friend}|\textit{Normal}) = 0.29$$

$$\mathbb{P}(\textit{Lunch}|\textit{Normal}) = 0.18$$

$$\mathbb{P}(\textit{Money}|\textit{Normal}) = 0.06$$

- This time, we get a new message “Lunch Money Money Money Money”

- $\mathbb{P}(\textit{Normal}) * \mathbb{P}(\textit{Lunch}|\textit{Normal}) * \mathbb{P}(\textit{Money}|\textit{Normal})^4 = 0.000002$

- $\mathbb{P}(\textit{Spam}) * \mathbb{P}(\textit{Lunch}|\textit{Spam}) * \mathbb{P}(\textit{Money}|\textit{Spam})^4 = 0$



$$\mathbb{P}(\textit{Dear}|\textit{Spam}) = 0.29$$

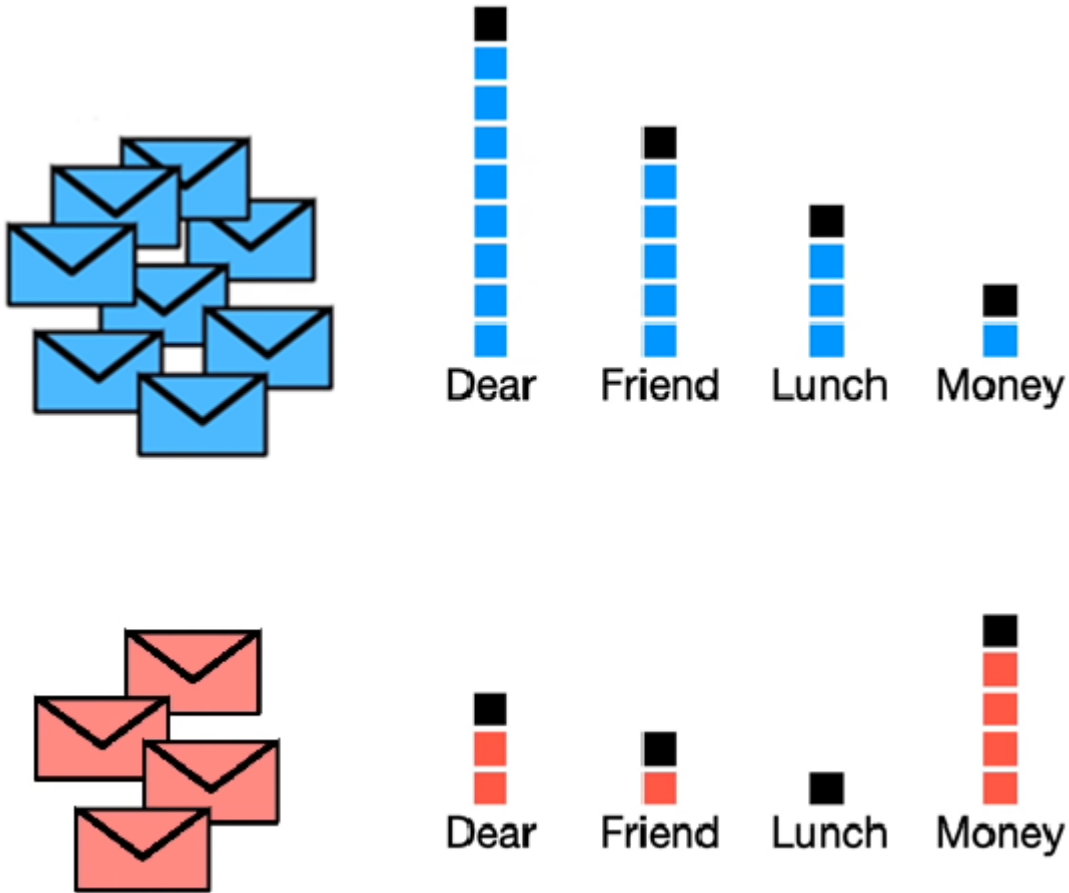
$$\mathbb{P}(\textit{Friend}|\textit{Spam}) = 0.14$$

$$\mathbb{P}(\textit{Lunch}|\textit{Spam}) = 0.00$$

$$\mathbb{P}(\textit{Money}|\textit{Spam}) = 0.57$$

- Because $\mathbb{P}(\textit{Lunch}|\textit{Spam}) = 0$

Toy Example: Naïve Bayes



- To work around this, we add one block to each of the words
- $\mathbb{P}(Lunch|Spam) = 1/11=0.09$

Toy Example: Naïve Bayes



$$\mathbb{P}(\text{Dear}|\text{Normal}) = 0.43$$

$$\mathbb{P}(\text{Friend}|\text{Normal}) = 0.29 \quad \bullet \quad \mathbb{P}(\text{Normal}) * \mathbb{P}(\text{Lunch}|\text{Normal}) * \mathbb{P}(\text{Money}|\text{Normal})^4 = 0.00001$$

$$\mathbb{P}(\text{Lunch}|\text{Normal}) = 0.19$$

$$\mathbb{P}(\text{Money}|\text{Normal}) = 0.10 \quad \bullet \quad \mathbb{P}(\text{Spam}) * \mathbb{P}(\text{Lunch}|\text{Spam}) * \mathbb{P}(\text{Money}|\text{Spam})^4 = 0.00122$$

- Calculate the scores again



$$\mathbb{P}(\text{Dear}|\text{Spam}) = 0.27$$

$$\mathbb{P}(\text{Friend}|\text{Spam}) = 0.18$$

$$\mathbb{P}(\text{Lunch}|\text{Spam}) = 0.09$$

$$\mathbb{P}(\text{Money}|\text{Spam}) = 0.45$$

- Now we see this is a Spam message

It's Naïve

- Naïve Bayes is naïve because it does not count the order of words, no grammar checking either
- In terms of machine learning, Naïve Bayes has a high bias since it ignores the relation among words
- And its variance is low
- As a very simple method, Naïve Bayes works surprisingly well in reality, at least for spam filter

Time to Play

- Let's see how to use Naïve Bayes in **python**!

